Solutions

1. (12 pts) Find the equation of the tangent plane to $f(x,y) = \sin(xy)$ at the point $(\frac{\pi}{4}, \frac{2}{3})$.

Answer: $z = \frac{1}{2} + \frac{\sqrt{3}}{3}(x - \pi/4) + \frac{\pi\sqrt{3}}{8}(y - 2/3)$

Show work here:

$f(\pi/4, 2/3) = \sin(\frac{\pi}{\cancel{4}_2} \cdot \frac{\cancel{2}}{3}) = \sin(\pi/6) = \frac{1}{2}$

$f_x = y\cos(xy) \Rightarrow f_x(\pi/4, 2/3) = \frac{2}{3}\cos(\pi/6) = \frac{\cancel{2}}{3} \cdot \frac{\sqrt{3}}{\cancel{2}} = \frac{\sqrt{3}}{3}$

$f_y = x\cos(xy) \Rightarrow f_y(\pi/4, 2/3) = \frac{\pi}{4}\cos(\pi/6) = \frac{\pi}{4} \cdot \frac{\sqrt{3}}{2} = \frac{\pi\sqrt{3}}{8}$

2. (9 pts) Estimate $\Delta f = f(1.02, 0.95) - f(1,1)$ assuming that $f(1,1) = 1,\ f_x(1,1) = 3,\ f_y(1,1) = -4$. Simplify your answer.

Answer: $\Delta f \approx 0.26$

Show work here:

$$\begin{aligned} f(1.02, 0.95) \approx L(1.02, 0.95) &= 1 + 3(1.02 - 1) - 4(0.95 - 1) \\ &= 1 + 3(0.02) - 4(-0.05) \\ &= 1 + 0.06 + 0.2 \\ &= 1.26 \end{aligned}$$

$\Delta f = f(1.02, 0.95) - f(1,1) \approx L(1.02, 0.95) - f(1,1) = 1.26 - 1 = 0.26$

or use

$\Delta f = f(x,y) - f(a,b) \approx f_x(a,b)\Delta x + f_y(a,b)\Delta y$

3. (12 pts) Let $f(x,y,z) = y^2 - 2xy - z^2$, $\mathbf{v} = \langle 3, 0, 4\rangle$, and $P = (1, 1, -1)$.

(3.1) a. A unit vector that points in the direction of the maximum rate of increase f at P is $\langle -1/\sqrt{2}, 0, 1/\sqrt{2} \rangle$ or $\langle -2/\sqrt{8}, 0, 2/\sqrt{8} \rangle$.

b. The maximum rate of increase of f at P is $2\sqrt{2}$ or $\sqrt{8}$.

Show all work:

$$\nabla f = \langle -2y, 2y - 2x, -2z \rangle$$

$$\nabla f_P = \nabla f(1,1,-1) = \langle -2, 0, 2 \rangle$$

$$\| \nabla f_P \| = \sqrt{(-2)^2 + 0^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

(3.2) Find the directional derivative of f in the direction of $\mathbf{v}$ at the point P. Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

(A) $\langle 0, 0, 0\rangle$	(E) $\langle 1, 1, -1\rangle$	(I) 2
(B) 5	(F) $2\sqrt{2}$	(J) $\langle 3, 0, 4\rangle$
(C) $\frac{1}{\sqrt{2}}$	(G) $\langle -2, 0, 2\rangle$	(K) $\left\langle -\frac{6}{5}, 0, \frac{8}{5}\right\rangle$
(D) $\left\langle -\frac{6}{25}, 0, \frac{8}{25}\right\rangle$	(H) $\frac{2}{25}$	(L) $\frac{2}{5}$

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 3, 0, 4\rangle}{\sqrt{9+0+16}} = \langle 3/5, 0, 4/5 \rangle$$

$$D_{\vec{u}} f(P) = \langle -2, 0, 2 \rangle \cdot \langle 3/5, 0, 4/5 \rangle$$

$$= -\frac{6}{5} + 0 + \frac{8}{5}$$

$$= \frac{2}{5}$$

4. (9 pts) Let $f(x,y)$ be such that $\nabla f(x,y) = \langle 4x^3y, x^4\rangle$, and let $x(s,t) = s^2$ and $y(s,t) = st^2$. Apply the multivariable calculus chain rule to calculate $\frac{\partial f}{\partial s}$. Write only your final answer in the box provided. Your final answer must be simplified and written in terms of s and t.

$$\frac{\partial f}{\partial s} = \boxed{9s^8t^2}$$

Show your work here:

$$\frac{\partial x}{\partial s} = 2s, \quad \frac{\partial y}{\partial s} = t^2$$

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

$$= 4x^3y(2s) + x^4(t^2)$$

$$= 4(s^2)^3(st^2)(2s) + (s^2)^4t^2$$

$$= 8s^8t^2 + s^8t^2$$

$$= 9s^8t^2$$

5. (6 pts) Use implicit differentiation to find $\frac{\partial z}{\partial x}$ for $z^3 + zx + x^2y = x + 2yz$. Fill in **one** bubble that corresponds to your answer. Questions with multiple answers selected will receive no credit. No work required.

(A) $z + 2xy - 1$

(B) $3z^2 + x - 2y$

(C) $\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(D) $-\frac{x^2}{3z^2 + x}$

(E) $-\frac{3z^2 + x - 2y}{z + 2xy - 1}$

(F) $x^2 - 2z$

(G) $-\frac{3z^2 + x - 2y}{x^2 - 2z}$

(H) $\frac{x^2 - 2z}{3z^2 + x - 2y}$

(I) $-\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(J) $-\frac{z + 2xy}{3z^2 + x}$

(K) $-3z^2 - x + 2y$

(L) $-\frac{x^2 - 2z}{3z^2 + x - 2y}$

$$F(x,y,z) = z^3 + zx + x^2y - x - 2yz = 0$$

$$F_x = z + 2xy - 1$$

$$F_z = 3z^2 + x - 2y$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{z + 2xy - 1}{3z^2 + x - 2y}$$

6. (10 points) Find the critical point(s) of $f(x,y) = 15x^2 - 2x^3 + 3y^2 + 6xy$. Write your final answer(s) on the line provided.

Critical point(s): $(0,0),\ (4,-4)$

Show work here:

$$\begin{cases} (1)\ f_x = 30x - 6x^2 + 6y = 0 \\ (2)\ f_y = 6y + 6x = 0 \Rightarrow 6y = -6x \Rightarrow y = -x \end{cases}$$

substitute into $y=-x$ (1) : $30x - 6x^2 - 6x = 0$

$\Rightarrow 24x - 6x^2 = 0$

$\Rightarrow 6x(4-x) = 0$

$\Rightarrow x=0,\ x=4$

when $x=0$ from (2) $y=0$, so $(0,0)$ is a critical pt

when $x=4$ from (2) $y=-4$, so $(4,-4)$ is a critical pt

7. (10 points) Let $P = (-4, 1)$ be a critical point of a function $f(x, y)$, and let $Q = (1, 2)$ be a critical point of a function $g(x, y)$. Assume that $f_{xx} = 2, f_{xy} = 0, f_{yy} = 4$ and $g_{xx} = 2, g_{xy} = -1, g_{yy} = 0$.

(7.1) Apply the second derivative test to classify P. Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

- (A) f has a local maximum at P.
- (B) f has a absolute maximum at P.
- (C) [filled] f has a local minimum at P.
- (D) f has a absolute minimum at P.
- (E) f has a saddle point at P.
- (F) Second derivative test is inconclusive at P.

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(P) = \begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} \; 2(4) - 0^2 = \underbrace{8 > 0}_{\text{must include}} \text{ and } f_{xx}(P) = \underbrace{2 > 0}_{\text{must include}} \;\cup$$

or $f_{yy}(P) = 4 > 0$

or $D(P) = f_{xx}(P) f_{yy}(P) - (f_{xy}(P))^2$

(7.2) Apply the second derivative test to classify Q. Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

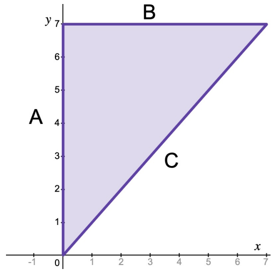
- (A) g has a local maximum at Q.
- (B) g has an absolute maximum at Q.
- (C) g has a local minimum at Q.
- (D) g has a absolute minimum at Q.
- (E) [filled] g has a saddle point at Q.
- (F) Second derivative test is inconclusive at Q.

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(Q) = \begin{vmatrix} 2 & -1 \\ -1 & 0 \end{vmatrix} = 2(0) - (-1)^2 = \underbrace{-1 < 0}_{\text{must include}}$$

8. (12 points) Let $f(x, y) = x^2 + y^2 - 6x - 8y$ on the domain D such that $x \geq 0$, $0 \leq y \leq 7$, and $y \geq x$ (see figure below).

(8.1) The maximum and minimum values of f on edge A have been provided. Find the maximum and minimum values of f on each of the remaining two edges of the triangle for the domain D. Complete the table with your final answers.



edge of triangle	A	B	C
max value of f	0	0	0
min value of f	-16	-16	$-49/2$

Show work below. You must provide justification for your answers for **each** remaining edge.

$B: y=7,\ 0 \leq x \leq 7: f(x,7) = x^2+49-6x-56 = x^2-6x-7$

Let $g(x) = x^2-6x-7$

$g'(x) = 2x-6 = 0 \Rightarrow x = 3$

$\underbrace{g(0)}_{=f(0,7)} = -7$, $\underbrace{g(3)}_{=f(3,7)} = 9-18-7 = -16$, $\underbrace{g(7)}_{=f(7,7)} = 49-42-7 = 0$

$C: y=x,\ 0 \leq x \leq 7: f(x,x) = x^2+x^2-6x-8x = 2x^2-14x$

Let $h(x) = 2x^2-14x$

$h'(x) = 4x-14 = 0 \Rightarrow x = 7/2$

$\underbrace{h(0)}_{=f(0,0)} = 0$, $\underbrace{h(7/2)}_{=f(7/2,7/2)} = 2(7/2)^2 - 14(7/2) = \frac{49}{2} - \frac{98}{2} = -49/2$

$h(7) = 2(7)^2 - 14(7) = 0 = f(7,7)$ (already found)

(8.2) The only critical point of the function f is $(3, 4)$. Determine the absolute maximum and absolute minimum of value of f on the domain D. Put your answers in the spaces provided.

The absolute max value of f is $\underline{\quad 0 \quad}$ and the absolute min value of f is $\underline{\quad -25 \quad}$.

Show any additional work here:

$f(3,4) = 3^2+4^2-6(3)-8(4)$

$= 9+16-18-32$

$= -25$

9. (10 points) Consider the problem of finding the points on the circle $x^2 + y^2 = 18$ at which the product xy is a maximum. Fill in the blanks below to set up this problem using the method of Lagrange multipliers.

(9.1) The function you need to maximize is $f(x, y) =$ xy

(9.2) The system you need to solve using Lagrange multipliers is a system with 3 (#) equations and 3 (#) unknowns.

(9.3) Using the method of Lagrange multipliers, write out the system of equations you need to solve for this problem. You **DO NOT** need to solve the system. Write your equations in the box below. (Do not just write the general equations.)

$$\begin{cases} y = \lambda(2x) \\ x = \lambda(2y) \\ \underbrace{x^2+y^2-18=0}_{\text{ok if } x^2+y^2=18} \end{cases}$$

or

$$\langle y, x\rangle = \lambda\langle 2x, 2y\rangle$$
$$x^2+y^2-18=0$$

Note: $\nabla f = \lambda \nabla g \Rightarrow \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$ must follow from what was given for $f(x,y)$ in 9.1 and the constraint equation in 9.3.

10. (10 points) For each true/false question, fill in the appropriate bubble. No work required.

(10.1) (T) (F) [F filled] If $f(x, y) = x \ln(x + y)$, then $(0, 0)$ is a critical point of f. f not defined at (0,0)

(10.2) (T) (F) [T filled] The gradient $\nabla F(a, b, c)$ is perpendicular to the tangent plane to the level surface $F(x, y, z) = k$ at the point $P = (a, b, c)$. $\nabla F(a,b,c)$ is the normal vector to the tangent plane at P.

(10.3) (T) (F) [T filled] For a function $f(x, y)$ and point $P = (a, b)$: $f_x(a, b) = D_{\mathbf{u}} f(a, b)$ when $\mathbf{u} = \langle 1, 0\rangle$.

(10.4) (T) (F) [F filled] If a function $f(x, y)$ has continuous third-order partial derivatives then $f_{xyy} = f_{xxy}$.

(10.5) (T) (F) [F filled] If $\nabla f(a, b) = \langle 0, 0\rangle$, then f has a local maximum or minimum at (a, b).